

ECON 108

PRINCIPLES OF ECONOMICS

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Perfect Competition

CHAPTER XII

CHAPTER OUTLINE

Competitive markets

Short-run profit maximization and profit

Short-run supply curve of the firm

Short-run market supply

Long-run market supply

COMPETITIVE MARKETS

Competitive Market

- In a competitive market firms are **price takers**
- A price taker has no control over the price it pays, or receives, in the market.

COMPETITIVE MARKETS

Perfectly Competitive Markets

◦ **Perfect competition** is a model describing competitive markets. Typical assumptions are:

- Consumers believe that all firms in the market sell *identical* (homogeneous) products.
- Large number of buyers and sellers.
- *Buyers and sellers know the prices* charged by firms.
- *Transaction costs* are low.
- Firms *freely enter and exit* the market.

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization

- Recall:

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

- Producers apply marginal thinking to maximize profit: when deciding whether to produce an additional unit or not firms weight additional revenue and additional cost

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization

- **Marginal revenue** is the change in total revenue divided by a change in the quantity sold.
- Measures the additional revenue obtained by selling one more unit of the product.

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization

- **Marginal cost** is the change in total costs due to a one-unit change in production rate

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization rule

- Profit maximization occurs when a firm expands output until marginal revenue is equal to marginal cost

$$MR = MC$$

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization rule

- If the firm chooses a quantity such that $MR > MC$, the cost of producing an additional unit is less than the revenue this additional unit brings in

INCREASE PRODUCTION

SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximization rule

- If the firm chooses a quantity such that $MR < MC$, the cost of the last unit produced is more than the revenue this last unit brings in

DECREASE PRODUCTION

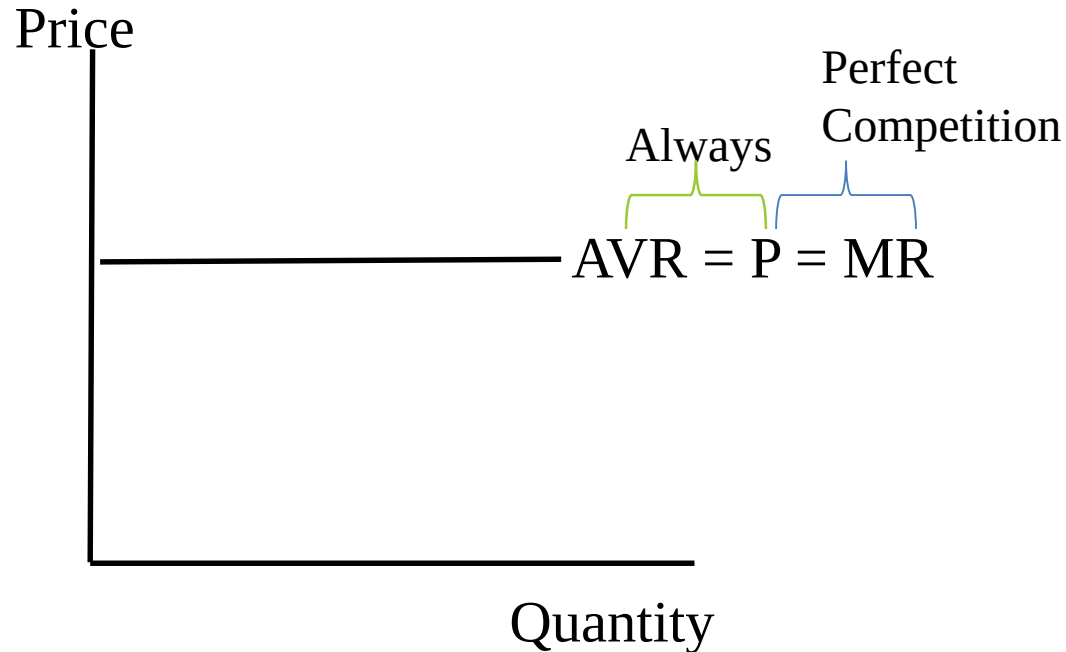
SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Relationship between price, average revenue, and marginal revenue

- **Average revenue** is equal to total revenue divided by the quantity sold. Average revenue is always equal to the price of the good.
- In a competitive market, since the firm is a price taker, marginal revenue is equal to the price.

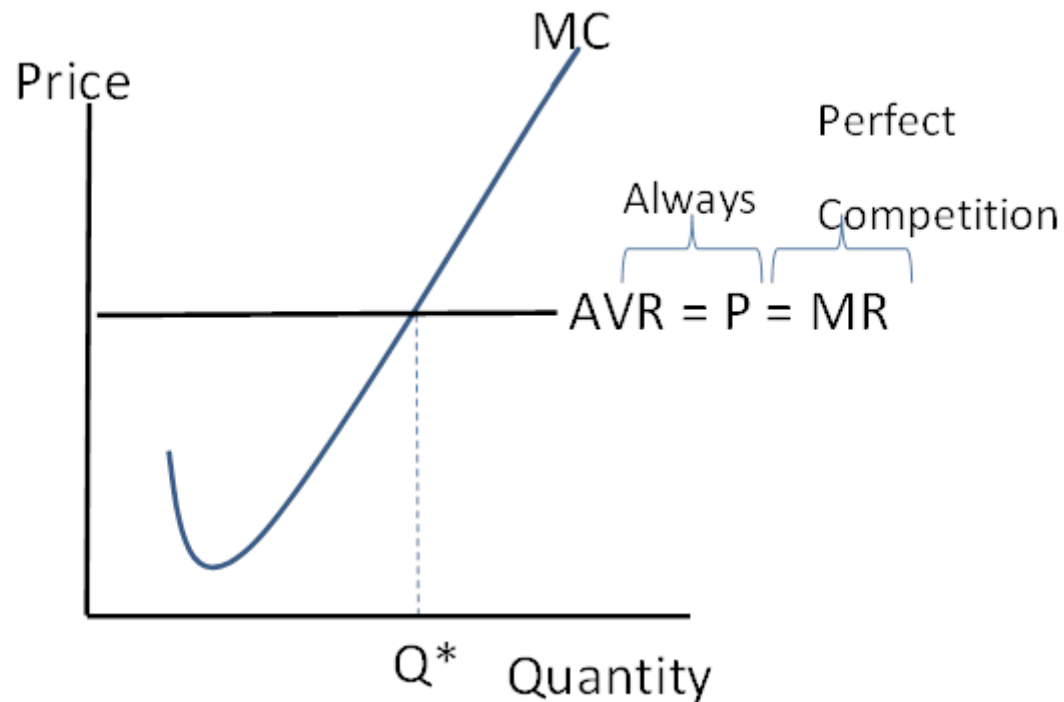
SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Relationship between price, average revenue, and marginal revenue



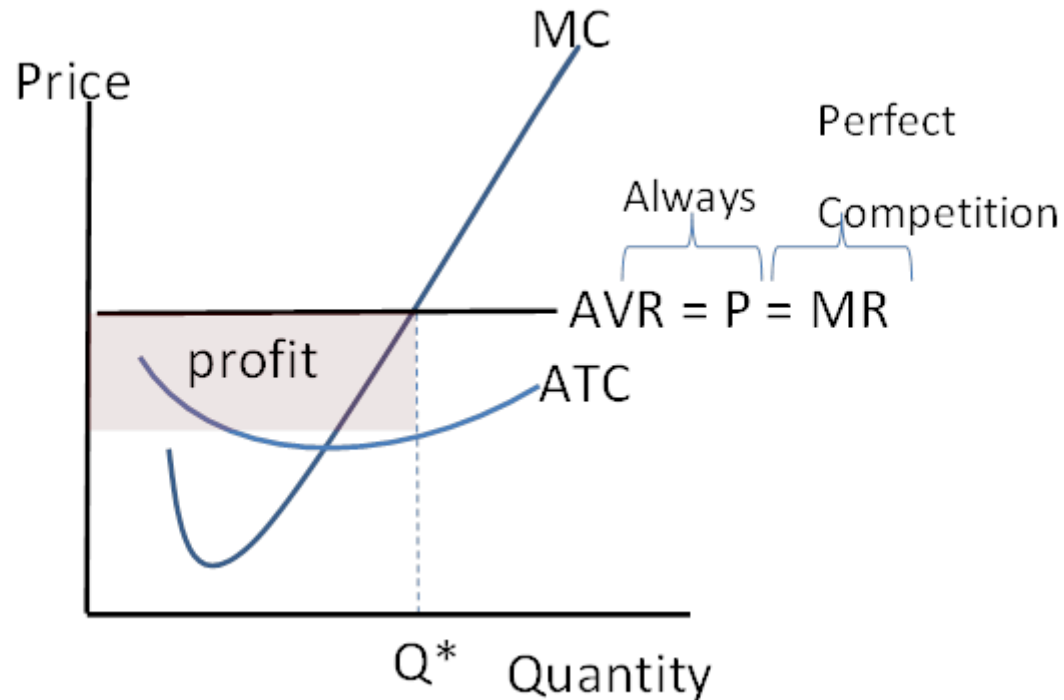
SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit maximizing output



SHORT-RUN PROFIT MAXIMIZATION AND PROFIT

Profit



SHORT-RUN SUPPLY CURVE

Short-Run Shut Down Decision

- The firm shuts down only if doing so reduces or eliminates its loss.
- The firm can gain by shutting down only if its revenue is less than its short run variable cost:

$$TR < TVC.$$

SHORT-RUN SUPPLY CURVE

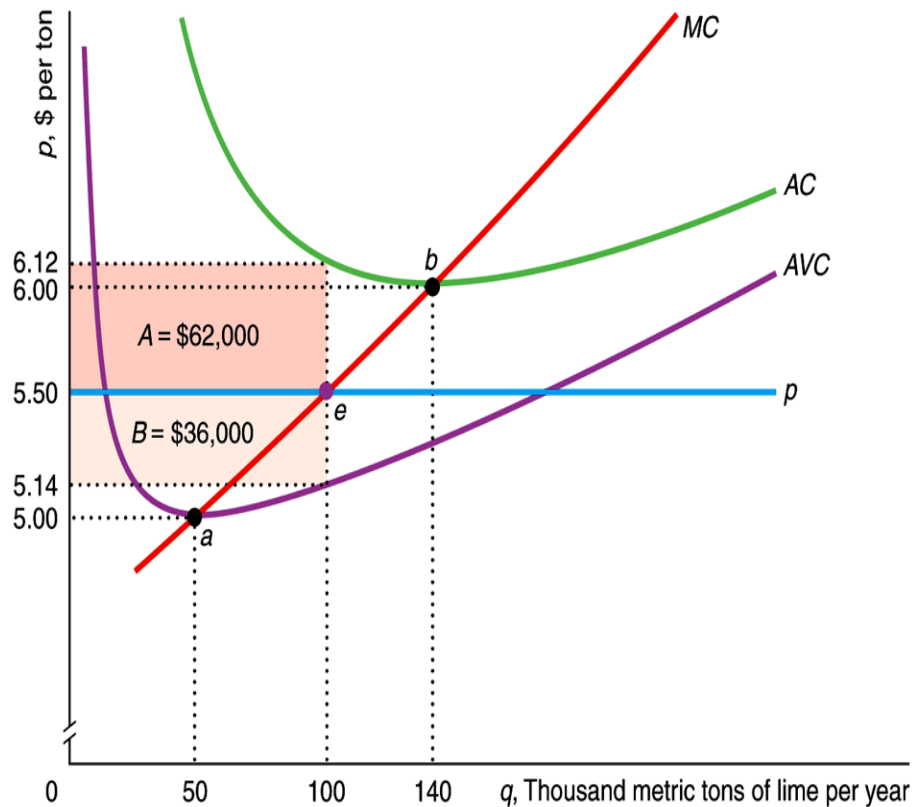
Short-Run Shut Down Decision

- Thus, a competitive firm shuts down if the market price is less than the minimum of its short-run average variable cost curve.

$$P < AVC$$

SHORT-RUN SUPPLY CURVE

Short-Run Shut Down Decision



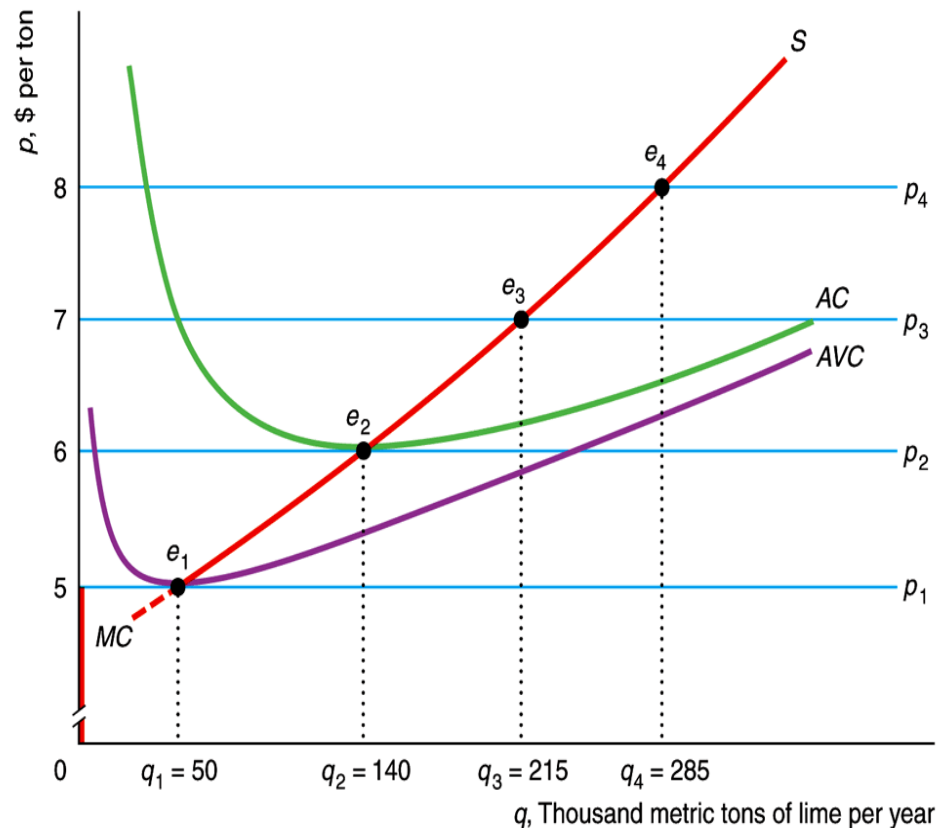
SHORT-RUN SUPPLY CURVE

Short-Run Firm Supply Curve

- A competitive firm shuts down if the market price is less than the minimum of its short-run average variable cost curve.
- The supply curve is the marginal cost curve when the price is above the minimum point on the average variable cost curve. Below that point, the firm will shut down and no supply exists.

SHORT-RUN SUPPLY CURVE

Short-Run Shutdown Decision



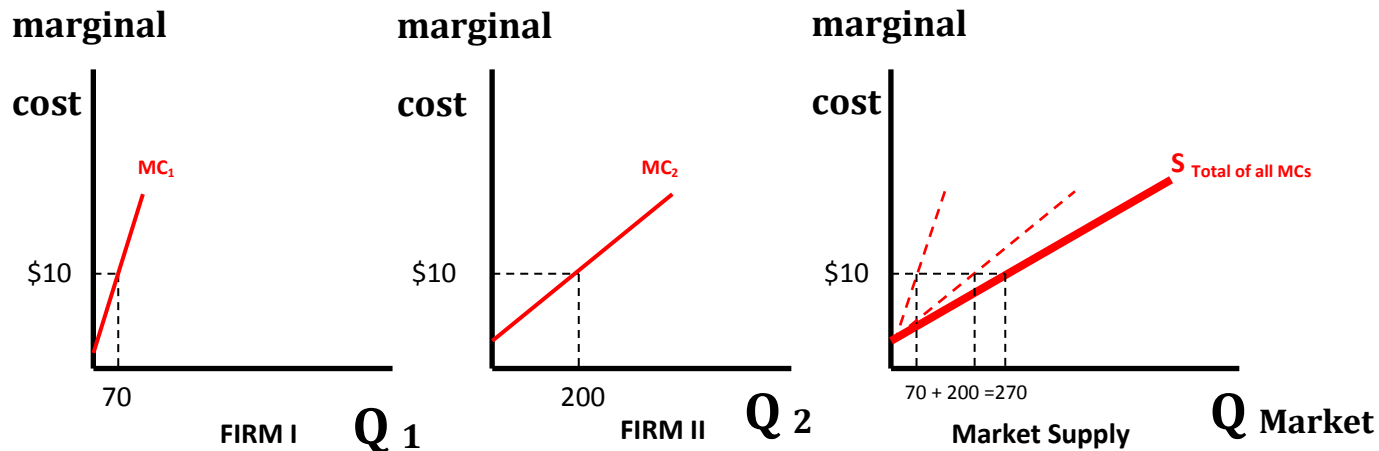
SHORT-RUN MARKET SUPPLY CURVE

Short-Run Market Supply Curve

- A competitive market consists of a large number of identical sellers.
- Since an individual firm's supply curve is equal to its marginal cost curve.
- If we add together all of the individual firm supply curves in a market we arrive at the **short-run market supply curve**

SHORT-RUN MARKET SUPPLY CURVE

Short-Run Market Supply Curve



The Long-Run Industry Situation: Exit and Entry

Profits and losses act as **signals** for resources to enter an industry or to leave an industry.

The Long-Run Industry Situation: Exit and Entry (cont'd)

Signals

- Signals are compact ways of conveying to economic decision makers information needed to make decisions.
- An effective signal not only conveys information but also provides an incentive to react appropriately.

The Long-Run Industry Situation: Exit and Entry (cont'd)

Exit and entry of firms:

- Economic profits:
 - Signal resources to enter the market
- Economic losses:
 - Signal resources to exit the market

The Long-Run Industry Situation: Exit and Entry (cont'd)

Allocation of capital and market signals:

- Price system allocates capital according to the relative expected rates of return on alternative investments.
- Investors and other suppliers of resources respond to market signals about their highest-valued opportunities.

The Long-Run Industry Situation: Exit and Entry (cont'd)

Tendency toward equilibrium (note that firms are adjusting all the time)

- At break-even, resources will not enter or exit the market.
- In competitive long-run equilibrium, firms will make zero economic profits.

The Long-Run Industry Situation: Exit and Entry (cont'd)

Long-run industry supply curve

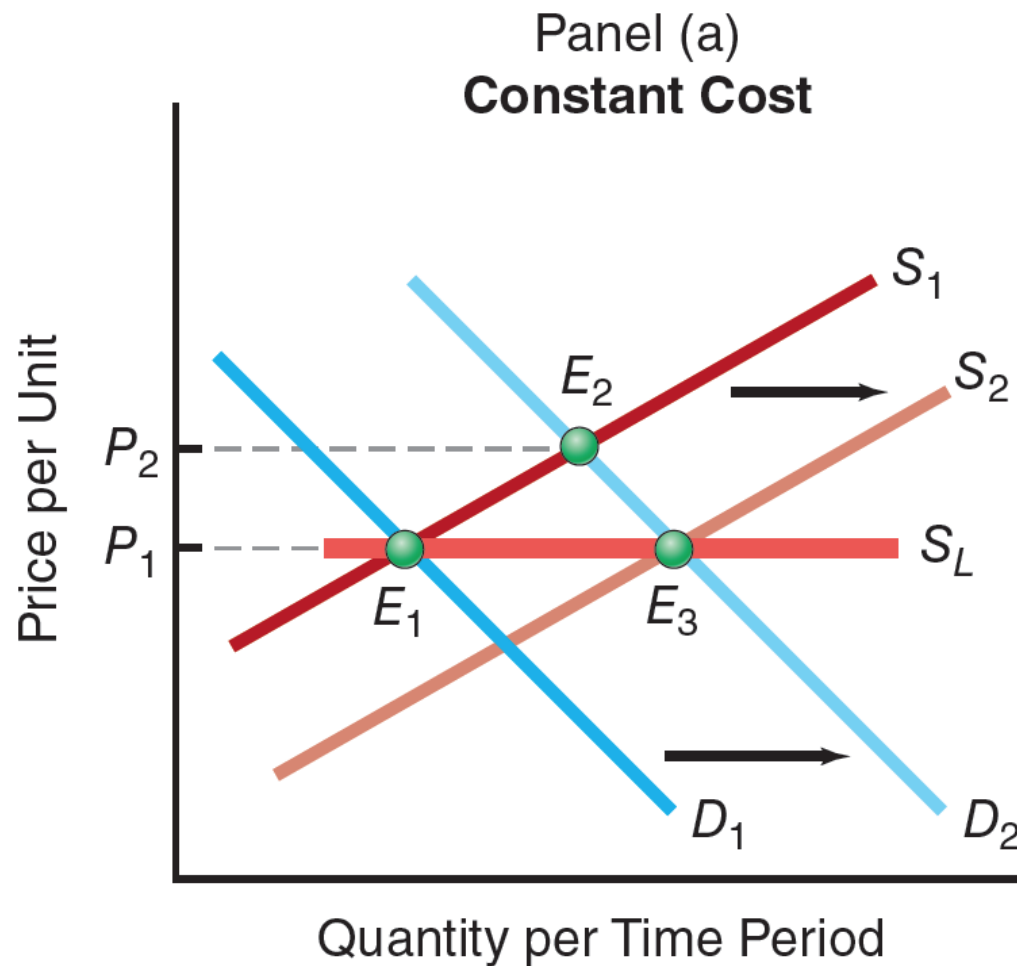
- A market supply curve showing the relationship between prices and quantities after firms have been allowed time to enter or exit from an industry, depending on whether there have been positive or negative economic profits

The Long-Run Industry Situation: Exit and Entry (cont'd)

Constant-cost industry

- In this type of industry, total output can be increased without an increase in long-run per-unit costs.
- Its long-run supply curve is horizontal.

Constant-Cost, Increasing-Cost, and Decreasing-Cost Industries, Panel (a)

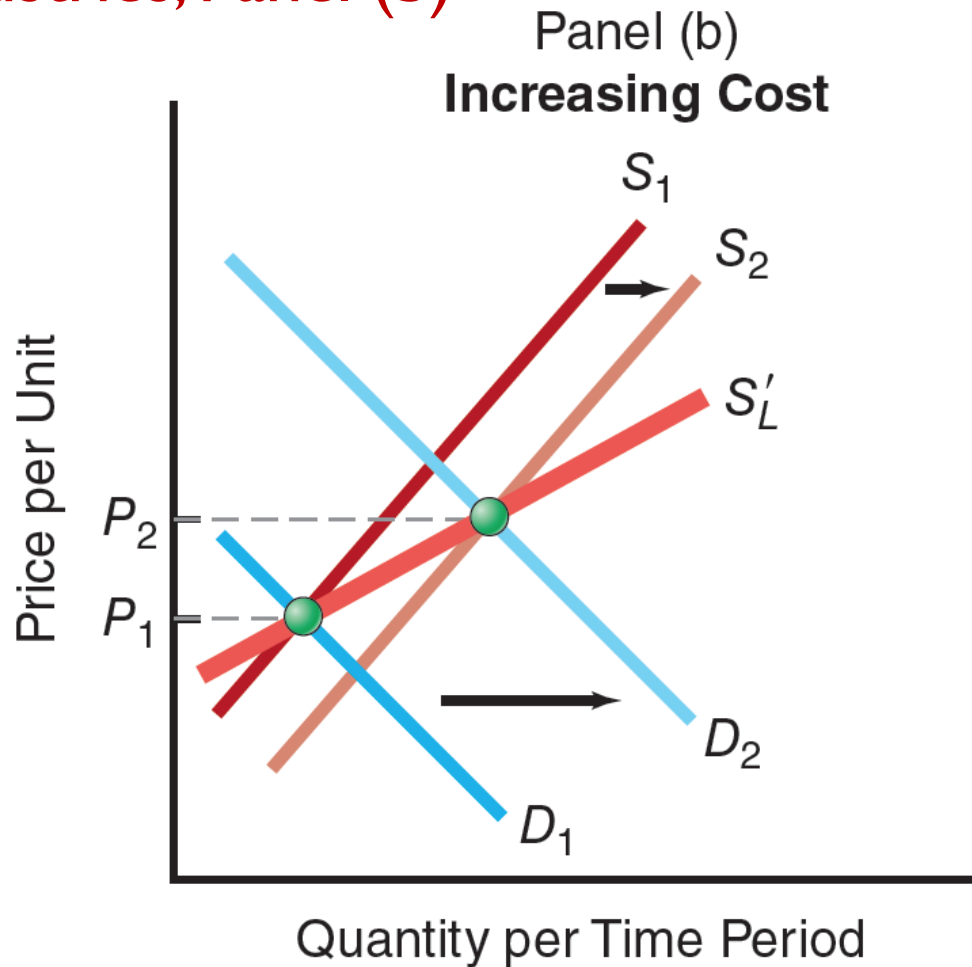


The Long-Run Industry Situation: Exit and Entry (cont'd)

Increasing-cost industry

- In this type of industry, an increase in industry output is accompanied by an increase in long-run per unit costs.
- Its long-run industry supply curve slopes upward.

Constant-Cost, Increasing-Cost, and Decreasing-Cost Industries, Panel (b)

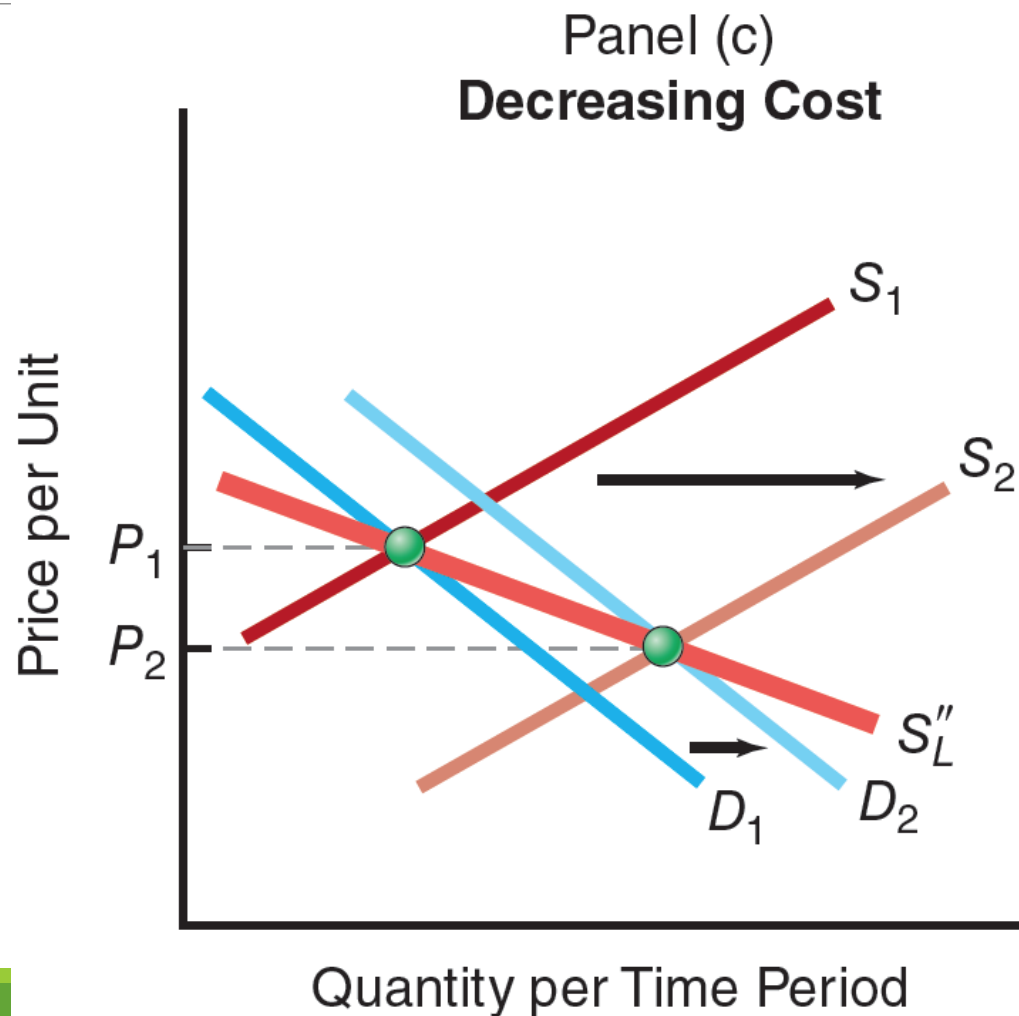


The Long-Run Industry Situation: Exit and Entry (cont'd)

Decreasing-cost industry

- In this type of industry, an increase in industry output leads to a reduction in long-run per-unit costs.
- Its long-run industry supply curve slopes downward.

Constant-Cost, Increasing-Cost, and Decreasing-Cost Industries, Panel (c)

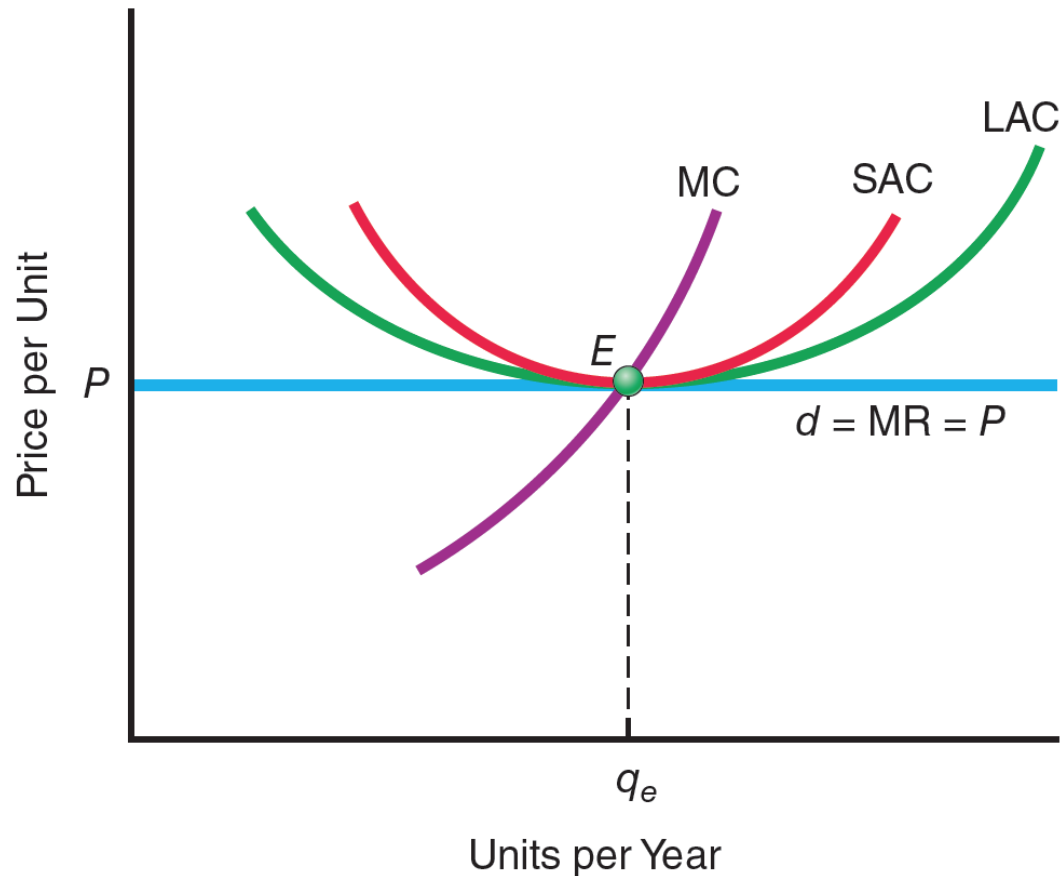


The Long-Run Industry Situation: Exit and Entry (cont'd)

In the long run, the firm can change the scale of its plant, adjusting its plant size in such a way that it has no further incentive to change; it will do so until profits are maximized.

In the long run, a competitive firm produces where price, marginal revenue, marginal cost, short-run minimum average cost, and long-run minimum average cost are equal.

Long-Run Firm Competitive Equilibrium



THE END!!!!!!